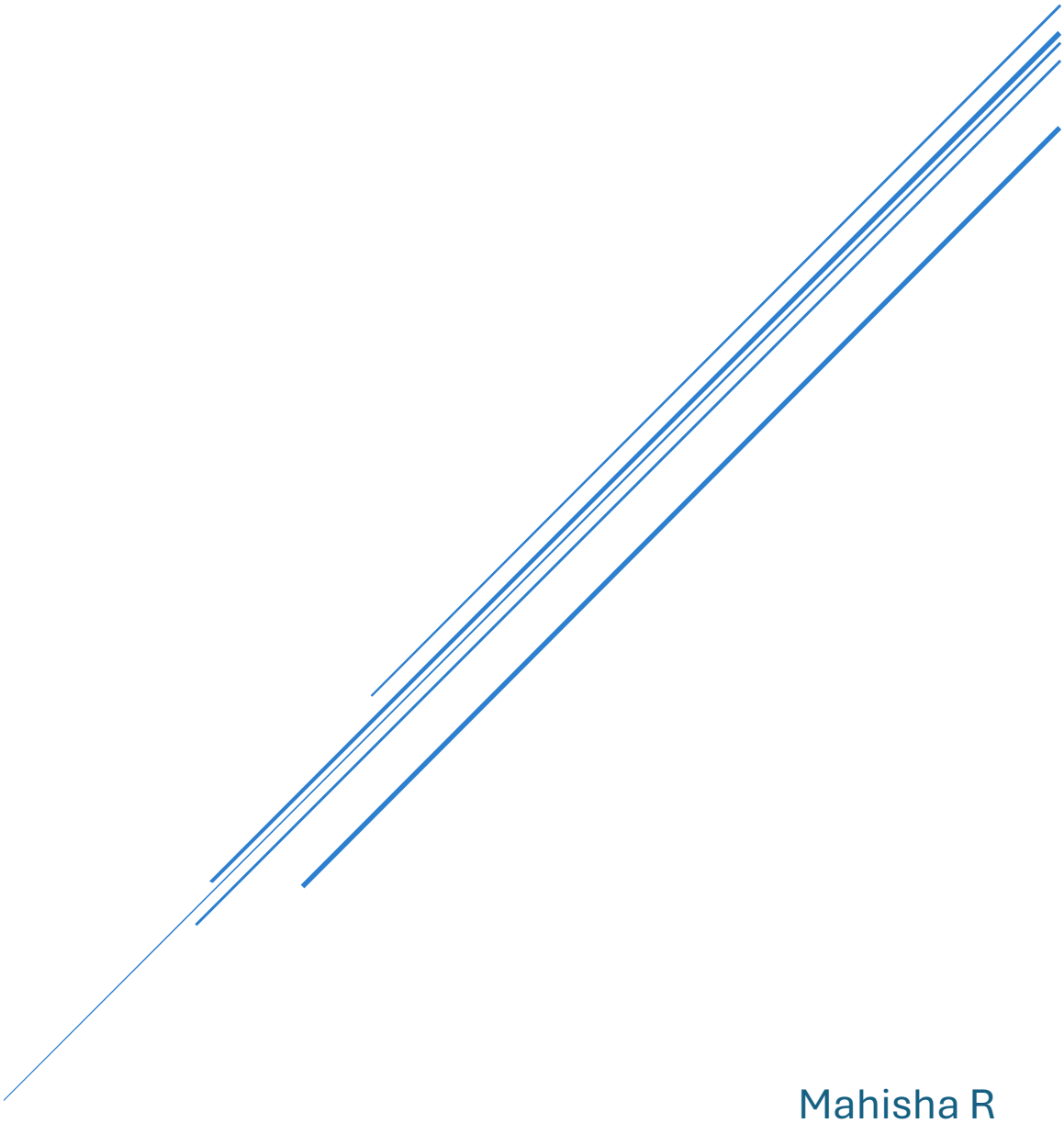
SQL PROJECT

Coded Project



Mahisha R

New Wheels Project

Business Context:

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many are now preferring pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Problem Statement:

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which concerns the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

Objective:

As a data scientist, you see that there is an array of questions that are being asked at the leadership level that needs to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

New Wheels Data Dictionary

---

shipper_id: Unique ID of the Shipper

shipper_name: Name of the Shipper

shipper_contact_details: Contact details of the Shipper

product_id: Unique ID of the Product

vehicle_maker: Vehicle Manufacturing company name

vehicle_model: Vehicle model name

vehicle_color: Color of the Vehicle

vehicle_model_year: Year of Manufacturing

vehicle_price: Price of the Vehicle

quantity: Ordered Quantity

customer_id: Unique ID of the customer

customer_name: Name of the customer

gender: Gender of the customer

job_title: Job Title of the customer

phone_number: Contact detail of the customer

email_address: Email address of the customer

city: Residing city of the customer

country: Residing country of the customer

state: Residing state of the customer

customer_address: Address of the customer

order_date: Date on which customer ordered the vehicle

order_id: Unique ID of the order

ship_date: Shipment Date

ship_mode: Shipping Mode/Class

shipping: Shipping Ways

postal_code: Postal Code of the customer

discount: Discount given to the customer for the particular order by credit card in percentage

credit_card_type: Credit Card Type

credit_card_number: Credit card number

customer_feedback: Feedback of the customer

quarter_number : Quarter Number

Business Overview:

	The whole year	Only the 4th quarter
Revenue in **million**	82.6	15.1
No of orders	1000	199
No of customers	994	199
Average Rating	3.14	2.40
Average Shipping Days	97.96	174.10

Total Revenue=82605431.10=82 million

Last quarter's revenue=15182375.04=15 million

BUSINESS QUESTIONS:

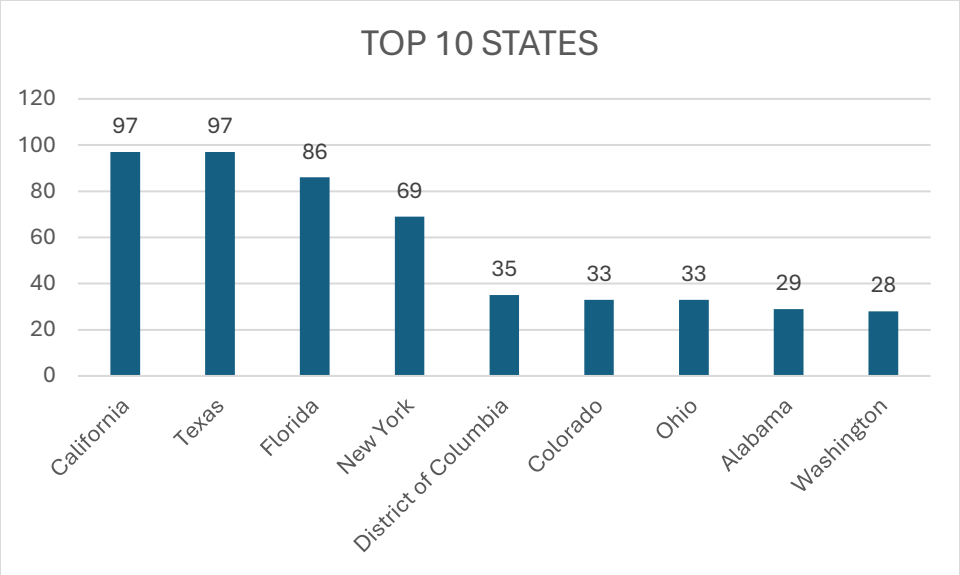
Question No 1:

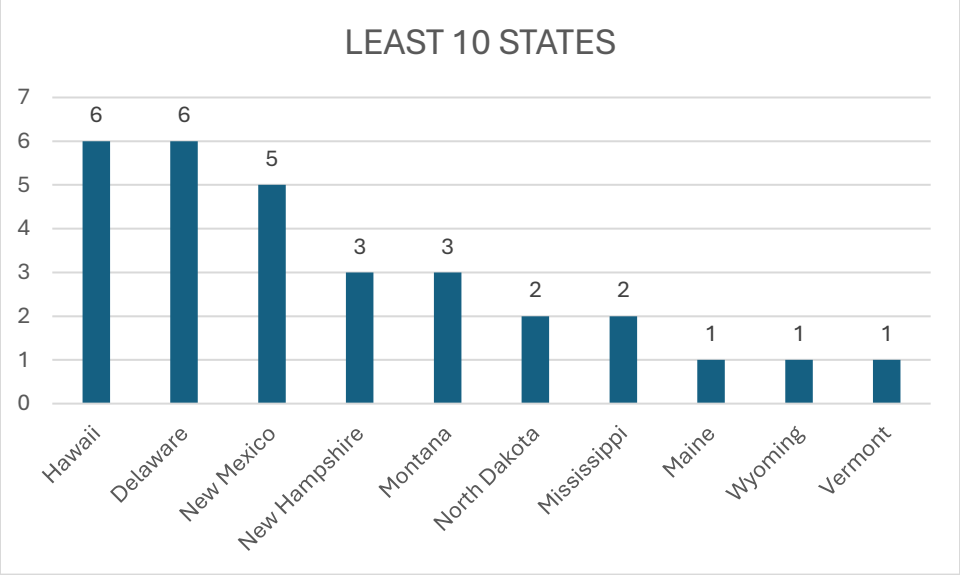
Find the total number of customers who have placed orders. What is the **distribution of the customers across states?**

```
SELECT
    state, COUNT(customer_id) AS customer_count
FROM
    customer_t
GROUP BY state
ORDER BY customer_count DESC;
```

Result Grid | Filter Rows:

state	customer_count
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26
Illinois	25
Pennsylvania	25
Virginia	24





- California, Texas, Florida and New York took the first four places with the highest customer count.
- Texas and California are the 2nd and 3rd largest states of USA.
- From 4th to 5th place, the count is reduced by 50%.
- Maine, Wyoming and Vermont took the least place with only one customer.

-- Question 2:

-- Which are the **top 5 vehicle makers preferred by the customers**?

```
-----------------------------------------
SELECT
    vehicle_maker, COUNT(DISTINCT (customer_id)) customer_count
FROM
    product_t
LEFT JOIN
    order_t ON order_t.product_id = product_t.product_id
GROUP BY vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
-------------------------------------
```

Result Grid | Filter Rows:

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

- Chevrolet is the most preferred vehicles with 83 vehicles, followed by Ford-63 and Toyota-52.

-- Question 3:

-- Which is the **most preferred vehicle maker in each state**?

```
WITH RankedVehicles AS (
  SELECT
    customer_t.state, product_t.vehicle_maker, COUNT(customer_t.customer_id) AS customer_count,
     RANK() OVER(partition by state order by COUNT(customer_t.customer_id) desc) AS ranking
  FROM
    customer_t
  JOIN order_t ON order_t.customer_id = customer_t.customer_id
  JOIN product_t ON order_t.product_id = product_t.product_id
  GROUP BY
    customer_t.state, product_t.vehicle_maker  )
SELECT
  state,
  vehicle_maker,
  customer_count
FROM
  RankedVehicles
WHERE
  ranking = 1;
  ------------------------------------------
```

Result Grid | Filter Rows:

state	vehicle_maker	customer_count
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Suzuki	1
Arkansas	Chevrolet	1
Arkansas	Pontiac	1
Arkansas	Volkswagen	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
California	Ford	6
California	Dodge	6
California	Audi	6
California	Nissan	6
California	Chevrolet	6

- Majority of the states has multiple preferred vehicles.
- It des not show any notable differences.

-- **Question 4:**

-- Find the overall average rating given by the customers. What is the **average rating in each quarter?**

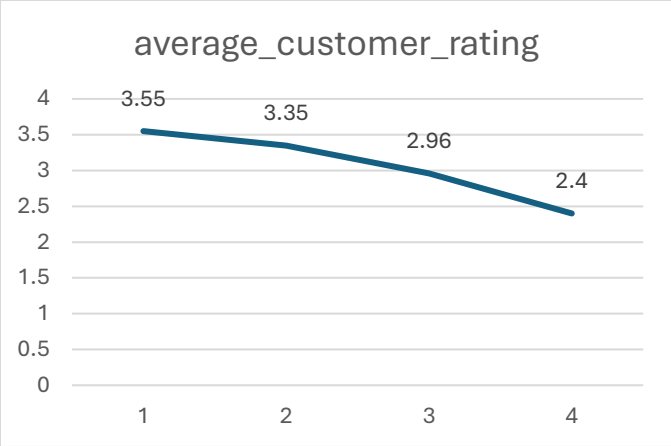
-- Consider the following mapping for ratings:

-- “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

```
WITH customer_rating AS(
    SELECT quarter_number,customer_feedback,
     CASE
            WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
     END AS rating
    FROM order_t)
SELECT
    quarter_number,
    ROUND(AVG(rating), 2) AS average_customer_rating
FROM
    customer_rating
GROUP BY quarter_number
ORDER BY quarter_number;
```

Result Grid | Filter Rows:

quarter_number	average_customer_rating
1	3.55
2	3.35
3	2.96
4	2.40



- Average Customer Rating is decreased gradually by over the quarters.
- From Q1 to Q4, Average Rating is dropped from 3.55bto 2.4.

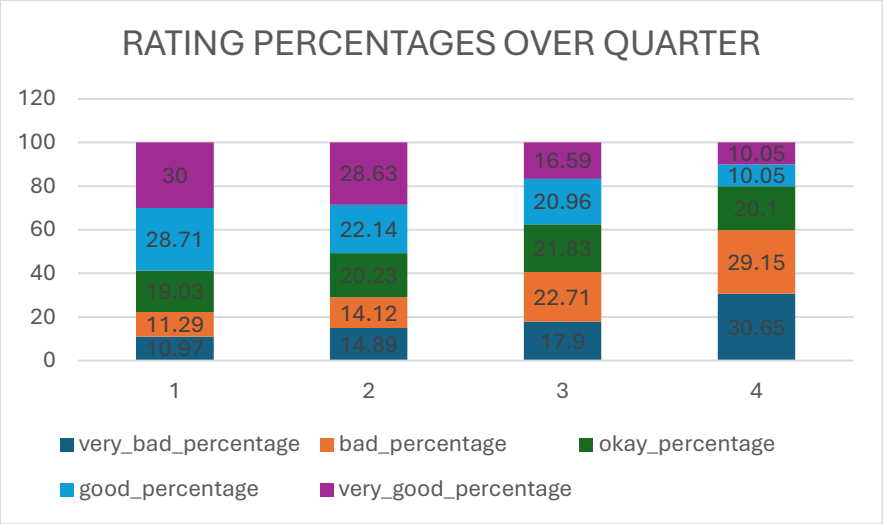
-- Question 5:

-- Find the **percentage distribution of feedback from the customers**. Are customers getting more dissatisfied over time?

```
WITH feedback_percentage AS (
  SELECT
    quarter_number,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) / COUNT(customer_feedback), 2) AS very_bad_percentage,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) / COUNT(customer_feedback), 2) AS bad_percentage,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) / COUNT(customer_feedback), 2) AS okay_percentage,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) / COUNT(customer_feedback), 2) AS good_percentage,
    ROUND(100.0 * SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) / COUNT(customer_feedback), 2) AS very_good_percentage
  FROM
    order_t
  GROUP BY
    quarter_number
  ORDER BY
    quarter_number )
SELECT
  quarter_number,
  very_bad_percentage,
  bad_percentage,
  okay_percentage,
  good_percentage,
  very_good_percentage
FROM
  feedback_percentage;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	very_bad_percentage	bad_percentage	okay_percentage	good_percentage	very_good_percentage
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

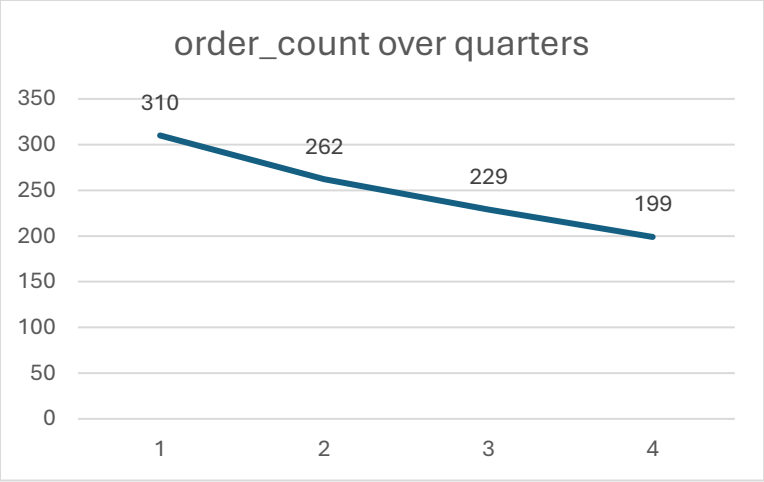


- Very_good_percentage and good_percentage is gradually decreased over the quarters.
- From Q1 to Q4 good_percentage dropped from 28.7% to 10.05%.
- From Q1 to Q4 very_good_percentage dropped from 30% to 10.05%.
- Ok_percentage stays stable between 19 to 20%.
- Very_bad_percentage and bad_percentage is gradually increased over the quarters.
- From Q1 to Q4 bad_percentage increased from 11.2% to 29.1%.
- From Q1 to Q4 very_bad_percentage increased from 10.9% to 30.6%.

-- Question 6:

-- What is the **trend of the number of orders by quarter**?

```
-------------------------------------
SELECT
  quarter_number,
  COUNT(order_id) AS order_count
FROM
  order_t
GROUP BY
  quarter_number
ORDER BY
  quarter_number;
```



- No of orders gradually decreased over the quarters.
- From Q1 to Q4, orders are decreased from 310 to 199.

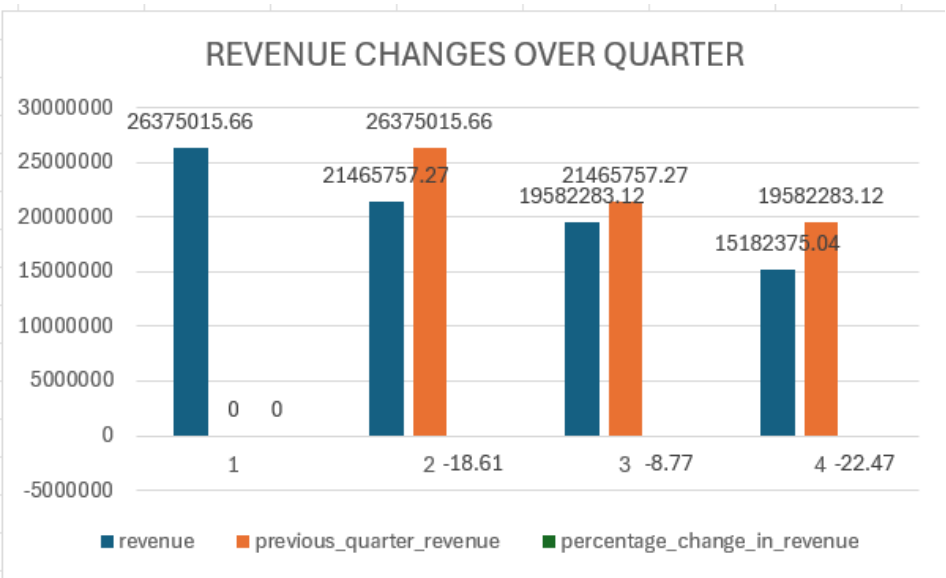
-- Question 7:

-- Calculate the **net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?**

```
-----------------------------------------------
WITH quarterly_revenue AS (
    SELECT
        quarter_number,
        ROUND(SUM(vehicle_price - ((discount / 100) * vehicle_price)), 2) AS revenue
    FROM
        order_t
    GROUP BY quarter_number
    ORDER BY quarter_number)
SELECT
    quarter_number,
    revenue,
    LAG(revenue) OVER (ORDER BY quarter_number) previous_quarter_revenue,
    ROUND(((revenue - LAG(revenue) OVER (ORDER BY quarter_number)) / LAG(revenue) OVER (ORDER BY quarter_number)) *100, 2) percentage_change_in_revenue
FROM
    quarterly_revenue;
-----------------------------------------------
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

quarter_number	revenue	previous_quarter_revenue	percentage_change_in_revenue
1	26375015.66	NULL	NULL
2	21465757.27	26375015.66	-18.61
3	19582283.12	21465757.27	-8.77
4	15182375.04	19582283.12	-22.47



- Each and every quarter the revenue is low compared to the previous quarters.
- Q4 shows the highest drop with -22% compared to previous quarter Q3.

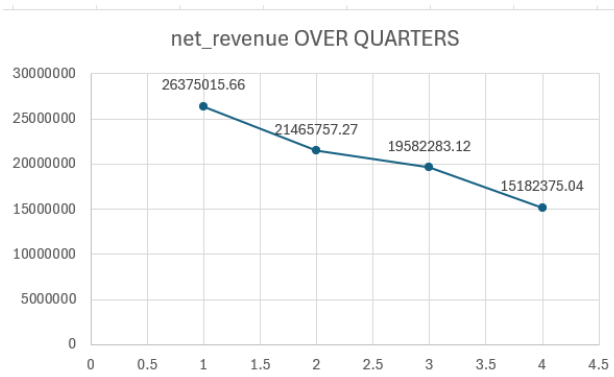
-- Question 8:

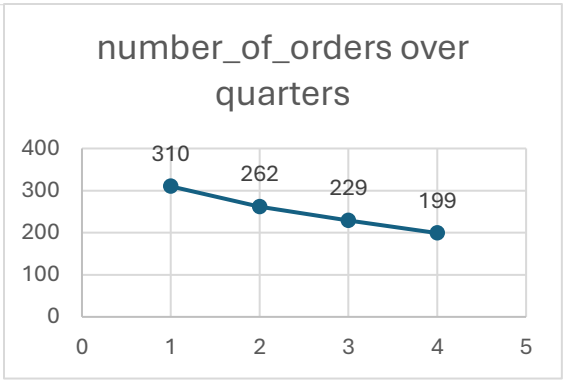
-- What is the trend of **net revenue and orders by quarters**?

```
SELECT quarter_number, COUNT(order_id) AS number_of_orders,
       ROUND(SUM(vehicle_price - ((discount / 100) * vehicle_price)), 2) AS net_revenue
FROM order_t
     GROUP BY quarter_number
     ORDER BY quarter_number;
```

Result Grid | Filter Rows: | Export

quarter_number	number_of_orders	net_revenue
1	310	26375015.66
2	262	21465757.27
3	229	19582283.12
4	199	15182375.04





- The no of orders gradually decreased over quarters and net revenue also gradually decreased.

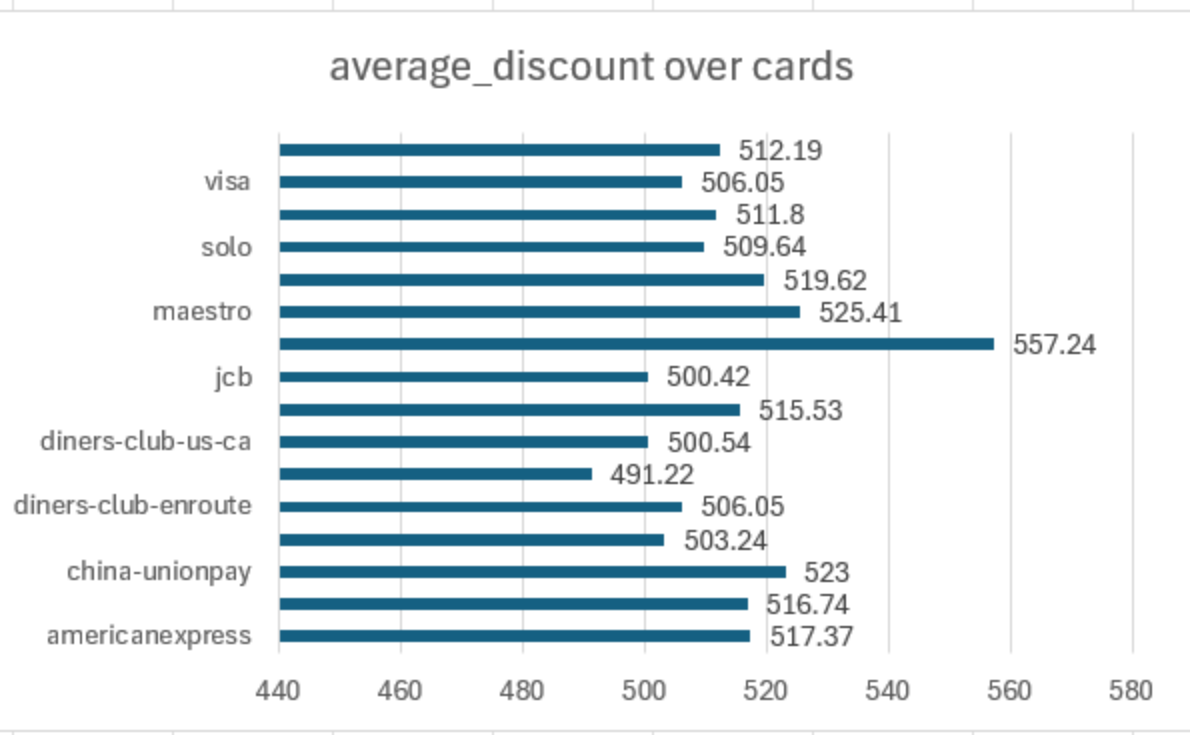
-- Question 9:

-- What is the **average discount offered for different types of credit cards?**

```
SELECT credit_card_type,
       ROUND(AVG ((discount/100)*vehicle_price),2) AS average_discount
FROM customer_t
     JOIN order_t ON order_t.customer_id = customer_t.customer_id
     GROUP BY credit_card_type
     ORDER BY credit_card_type;
     ---------------------------------
```

Result Grid | Filter Rows:

credit_card_type	average_discount
americanexpress	517.37
bankcard	516.74
china-unionpay	523.00
diners-club-carte-blanche	503.24
diners-club-enroute	506.05
diners-club-international	491.22
diners-club-us-ca	500.54
instapayment	515.53
jcb	500.42
laser	557.24
maestro	525.41
mastercard	519.62
solo	509.64
switch	511.80
visa	506.05
visa-electron	512.19



- Laser credit cards provided the highest discount with $557.24.
- Diners club international provided the least discount with $491.22.
- Notable differences among credit cards are not available in this data.

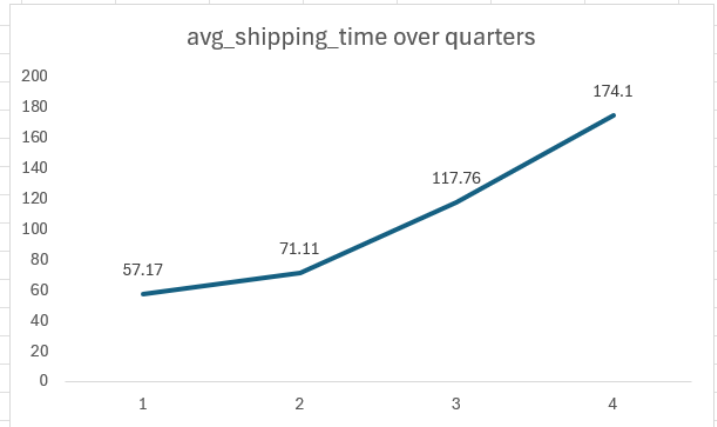
-- Question 10:

-- What is the **average time taken to ship the placed orders for each quarter**?

```
SELECT
    quarter_number,
    ROUND(AVG(DATEDIFF(ship_date,order_date)), 2) AS avg_shipping_time
FROM
    order_t
GROUP BY
    quarter_number
ORDER BY
    quarter_number;
```

Result Grid | Filter Rows:

quarter_number	avg_shipping_time
1	57.17
2	71.11
3	117.76
4	174.10



- Average shipping time is steadily increased over quarters.
- Q1 to Q4, Shipping time is drastically increased from 57 to 174 days. It is more than 150%.

Insights and Recommendations:

- Q1 to Q4, Shipping time is drastically increased from 57 to 174 days. It is more than 150%.
- Following the shipping delay time increased, average of consumer_rating is decreased from 3.5 to 2.4 over Q1 to Q4. It leads to revenue loss and no of orders also decreased.
- Especially in Q3 to, Q4 revenue is decreased from 19.5 to 15.1 million.
- Q1 to Q4, net revenue is decreased from 26 million to15 million. It is 22% less compared to Q3.
- Q1 to Q4, the no of orders is reduced from 310 to 199.
- The company is strongly recommended to decrease the shipping delay , to reduce the customer dissatisfaction.